Fpt University

TRƯỜNG ĐẠI HỌC FPT

# CAPSTONE PROJECT REPORT

## Report 5 – Software Testing Documentation


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# V. Software Testing Documentation

## 1. Scope of Testing

| Level | In-Charge | Inputs/Time | Focus | Acceptance Criteria |
|---|---|---|---|---|
| Unit Testing | Developers | Individual modules during development. | Business logic, validation and edge cases. | All unit tests pass; no critical bugs. |
| Integration Testing | All Members | After module integration. | APIs, data flow and external services. | Modules integrate correctly; no major errors. |
| Interface Testing | All Members | After UI implementation. | Role-based navigation, forms and messages. | UI works correctly; no broken navigation. |
| End-to-End Testing | All Members | Before release, after integration. | Onboarding, ingestion, detection, investigation and subscription workflows. | Core flows meet agreed requirements; no unresolved critical defects. |

# 2. Test Strategy

## 2.1 Testing Types

| Type | Focus |
|---|---|
| Functional | Verify approved workflows, input validation and expected results. |
| Security and Authorization | Check role permissions, tenant isolation and sensitive-data protection. |
| Reliability | Check duplicates, data preservation, retries and AI fallback. |
| Usability | Check clear navigation, messages, alerts and investigation information. |
| Regression | Rerun affected tests after changes and defect fixes. |

## 2.2 Test Levels

| Level | Approach |
|---|---|
| Unit | Test isolated logic with normal, invalid and boundary inputs. |
| Integration | Test module/API interactions with a test database and controlled external services. |
| Interface | Manually check role-based screens, navigation and validation. |
| End-to-End | Run complete business journeys on a known test build; record actual results and evidence. |

## 2.3 Supporting Tools

| Tool | Purpose |
|---|---|
| JUnit 5 / Maven | Run backend unit and integration tests. |
| ArchUnit / Testcontainers | Check module boundaries and database integration. |
| API client / Browser developer tools | Check API requests, UI behavior and workflow results. |
| AI test double / Payment sandbox / Notification sink | Control external responses and failure scenarios. |
| Excel / Git | Maintain test cases, results, evidence links and synthetic test data. |

# 3. Test Plan

## 3.1 Human Resources

| No. | Full Name | Student Code | Role | Responsibilities |
|---|---|---|---|---|
| 1 | Lê Hồng Đức | SE182106 | Leader | Planning, assign feature groups to test. Write tests and test reports. |
| 2 | Huỳnh Khả Tú | SE182134 | Member | Write tests and test reports. |
| 3 | Nguyễn Minh Huy | SE181524 | Member | Write tests and test reports. |
| 4 | Nguyễn Thị Diệu Vi | SE181505 | Member | Write tests and test reports. |

## 3.2 Test Environment

| Component | Planned Setup | Version / Details |
|---|---|---|
| Test Build | SME-FraudGuard test deployment | [Build / Commit] |
| Operating System | Test machine / server | [OS / Version] |
| Browser | Browser for interface and end-to-end tests | [Browser / Version] |
| Frontend | Next.js / React / Node.js / npm | [Versions] |
| Backend | Java / Spring Boot / Maven | [Versions] |
| Database | Isolated PostgreSQL test database | [Version] |
| AI Service | Controlled Normal, Anomaly and failure responses | [Model / Service Version] |
| External Services | Payment sandbox; notification sink | [Provider / Version] |
| Test Data | Synthetic data; separate workspaces A and B | [Dataset Version] |
| Environment Record | Deployment address and execution date | [Test URL]; [dd/mm/yyyy] |

## 3.3 Test Milestones

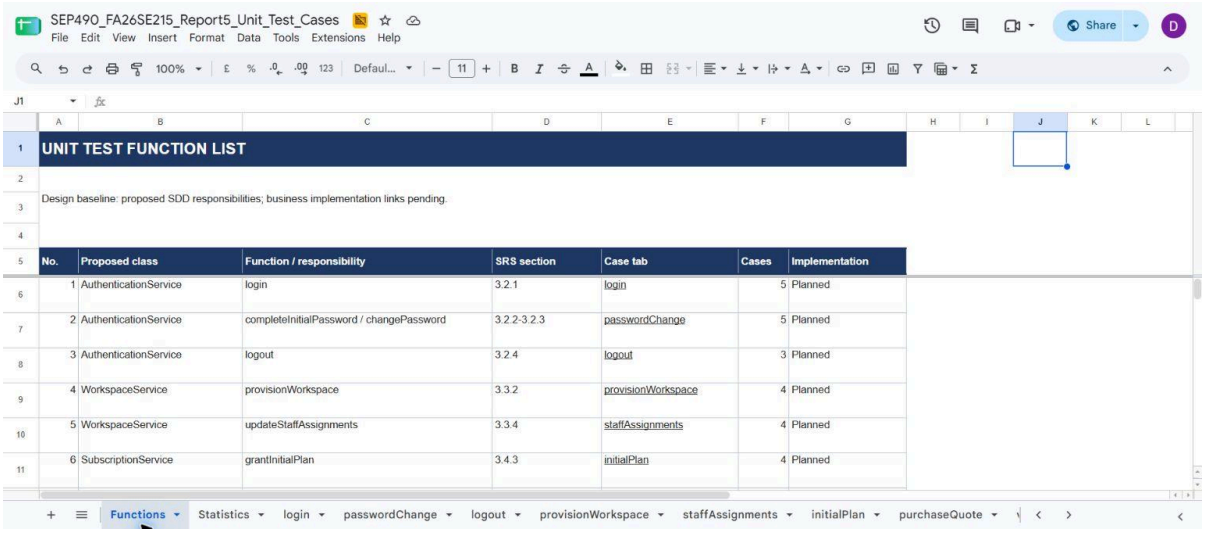
| Milestone | Start Date | End Date | Deliverable |
|---|---|---|---|
| Test Preparation | [dd/mm/yyyy] | [dd/mm/yyyy] | Test cases, synthetic data and test environment. |
| Unit Testing | [dd/mm/yyyy] | [dd/mm/yyyy] | Unit test results for completed modules. |
| Integration and Interface Testing | [dd/mm/yyyy] | [dd/mm/yyyy] | Integration/UI results and defect records. |
| End-to-End Testing | [dd/mm/yyyy] | [dd/mm/yyyy] | Core workflow results and evidence. |
| Regression and Reporting | [dd/mm/yyyy] | [dd/mm/yyyy] | Retest results and consolidated test report. |

# 4. Test Cases

## 4.1 Unit Test Cases

SEP490_FA26SE215_Report5_Unit_Test_Cases

164 designed case specifications in 33 function matrices. Class/function references follow the proposed SDD and will be aligned with implemented code. All cases are Not Run.



UNIT TEST FUNCTION LIST

Design baseline: proposed SDD responsibilities; business implementation links pending.

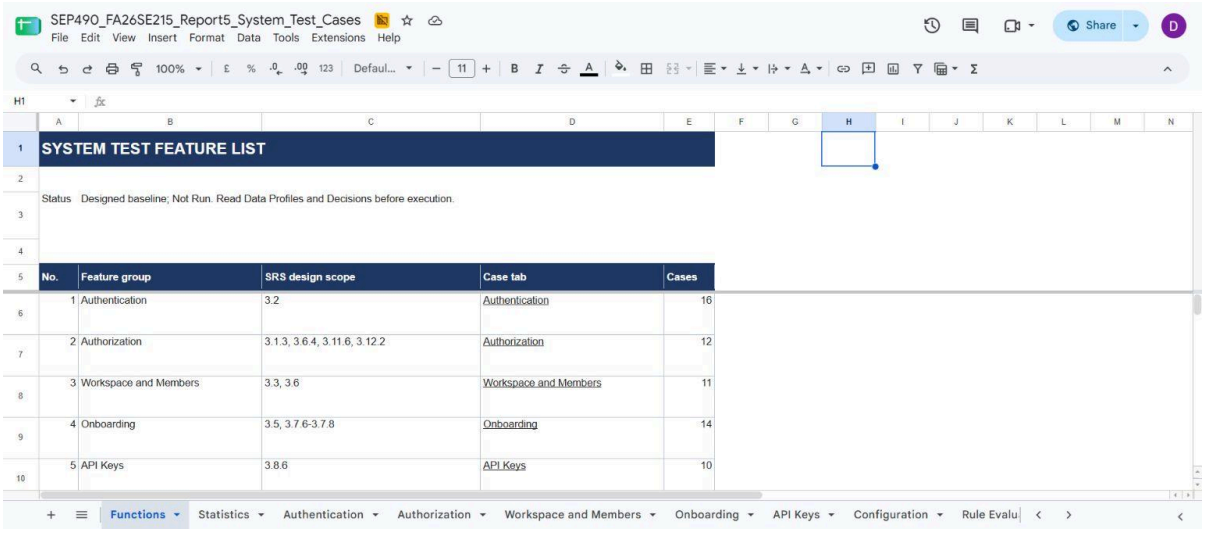
| No. | Proposed class | Function / responsibility | SRS section | Case tab | Cases | Implementation |
|---|---|---|---|---|---|---|
| 1 | AuthenticationService | login | 3.2.1 | login | 5 | Planned |
| 2 | AuthenticationService | completeInitialPassword / changePassword | 3.2.2-3.2.3 | passwordChange | 5 | Planned |
| 3 | AuthenticationService | logout | 3.2.4 | logout | 3 | Planned |
| 4 | WorkspaceService | provisionWorkspace | 3.3.2 | provisionWorkspace | 4 | Planned |
| 5 | WorkspaceService | updateStaffAssignments | 3.3.4 | staffAssignments | 4 | Planned |
| 6 | SubscriptionService | grantInitialPlan | 3.4.3 | initialPlan | 4 | Planned |

*Figure 5.4.1. Unit Test Case Catalogue (excerpt).*

### 4.2 System Test Cases

[SEP490_FA26SE215_Report5_System_Test_Cases]

191 designed cases in 18 feature groups. Detailed procedures, expected results and test-data profiles are maintained in the linked workbook. All cases are Not Run.



SEP490_FA26SE215_Report5_System_Test_Cases

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**SYSTEM TEST FEATURE LIST**

Status Designed baseline; Not Run. Read Data Profiles and Decisions before execution.

| No. | Feature group | SRS design scope | Case tab | Cases |
|---|---|---|---|---|
| 1 | Authentication | 3.2 | Authentication | 16 |
| 2 | Authorization | 3.1.3, 3.6.4, 3.11.6, 3.12.2 | Authorization | 12 |
| 3 | Workspace and Members | 3.3, 3.6 | Workspace and Members | 11 |
| 4 | Onboarding | 3.5, 3.7.6-3.7.8 | Onboarding | 14 |
| 5 | API Keys | 3.8.6 | API Keys | 10 |

Functions Statistics Authentication Authorization Workspace and Members Onboarding API Keys Configuration Rule Evalu

*Figure 5.4.2. System Test Case Catalogue (excerpt).*

# 5. Test Reports

### 5.1 Unit Test Report

### 5.2 System Test Report